

Dear Editors,

I am pleased to submit “**The near-wall barrier to parameter interpolation on AirFRANS is a coordinate artifact**” for consideration as an original research article in the *Journal of Computational Science*.

What the paper reports. Machine-learning surrogates for simulation are almost universally scored by resampling their predictions onto a uniform grid. We measure what that resampling costs, and the measurement changes the answer. On a standard external-aerodynamics benchmark, kernel interpolation over seven scalars parsed from the case file name — no network, no flow-field learning at all — gives $8.4\times$ lower volume-pressure error than a Transolver trained on the same data on an area-uniform 128^2 raster. Re-weighting the *identical* predictions by the dataset’s own node measure takes that ratio to $0.44\times$, and scoring per node on the native point cloud, with no rasterisation anywhere, takes it to $0.21\times$. One pair of predictions, three measures in current use, and the ranking reverses.

At native resolution the interpolator loses on every channel, and inside the first 0.005 chord the exact least-squares bound over the whole family — what *any* weighting of the 800 training fields can achieve — is $25\times$ worse than the surrogate on average and worse on every one of the 200 test cases. Read the usual way, that is where the paper would have stopped: a surrogate earns the boundary layer, and no interpolation reaches it.

We then measured why, and the explanation removed the barrier. Inside that band a query lands $7.8\times$ further from the training case’s wall than from its own, and 35% of the kernel’s weight falls *inside* the training airfoil. That is a statement about coordinates, not about representation, so it makes a prediction: pose the same family in a wall-following frame and the barrier should go. It does. With nothing retuned and the same 800 fields, the identical bound falls from $25\times$ to **$0.0044\times$** , **below the surrogate on all 200 cases**, while the physical-coordinate bound rescored on exactly the same nodes still reads $25.21\times$. The near-wall barrier our own native comparison measured is an artifact of the coordinate system the interpolation family was posed in.

We are careful about what that does and does not claim, and the manuscript is explicit about it. The finding rests on the *bound*, which is unanimous: paired per case, its worst ratio is $0.018\times$. The deployed estimator in the same frame improves $112\times$ on the case-mean but still trails the surrogate there ($1.71\times$); paired per case its median is $0.83\times$, below the surrogate on 115 of 200 cases. We therefore claim that no member of this family is barred from the wall by representation, and **not** that parameter interpolation beats a trained surrogate there. On pressure, for which no rule was registered, there was never a barrier — the physical bound already beats the surrogate by $974\times$ — and the wall-following frame makes it worse on 185 of 200 cases; the manuscript reports that reading as post hoc. The frame also has a measured cost: a wall-normal coordinate does not exist on 4.98% of near-wall nodes (1.36% of the verdict band), at the trailing edge and the leading-edge stagnation point, where distinct points project onto a single surface vertex.

The scoring raster cannot see any of this. Its own round-trip error at the native nodes exceeds the surrogate’s per-node error by $413\times$ pooled and $495\times$ near the wall. We state that symmetrically, because it is not a concession about one arm: it is a limit on what a 128^2 field metric can establish about *any* method on this benchmark, ours included, and it is why the head-to-head had to be re-run at native resolution rather than refined in place.

Why this belongs in the Journal of Computational Science. The contribution is an evaluation result — the object of study is the measurement protocol, not a new architecture — and this journal has recently and repeatedly published exactly that genre:

- *When simpler models win: a large-scale computational benchmark of lexical and transformer NLP pipelines for predicting medication effectiveness* (2026-08-27)
- *Exploring the limitations of transformer models for metocean forecasting* (2026-06-03)
- *Accuracy vs efficiency: benchmarking graph neural networks on edge GPU hardware* (2026-07-30)
- *Benchmarking atom-level explainability against pharmacophore-computed labels in molecular machine learning* (2026-07-08)

Two of these are limitations-of-a-model-class papers and two are head-to-head benchmarks; one is in a geophysical-fluids forecasting domain. The fit is to the journal’s demonstrated interest in whether a computational result’s reported number means what it appears to mean, which is precisely the question this paper asks of a widely used benchmark.

Where the paper is most exposed. We would rather state this than have it found.

- *The baseline estimator is not new.* The interpolator is kernel ridge regression over a seven-dimensional parameter space — response-surface methodology, and decades old. The paper says so in those words. It is deliberately not a contribution: its role is to be the cheapest defensible thing that could occupy a leaderboard position, and the finding is that on one published measure it does, which is a statement about the measure. A reader who wants a novel estimator will not find one here.
- *Scope is one benchmark, two dimensions, one learned architecture.* We do not claim the ratios transfer. What we argue transfers is the diagnosis — that a wall-clustered unstructured mesh scored on a uniform raster puts most of its nodes in a small fraction of the scoring area — and that is a property of the discretisation, checkable anywhere, not a property of this dataset. The paper states the limitation rather than hedging it.
- *This paper twice refuted its own headline, in public, using rules it had committed in advance.* The $8.4\times$ advantage was once the lead claim; the pre-registered rule that governed it returned ARTIFACT and the manuscript withdraws it in those words. The near-wall representational claim replaced it, and the body-fitted arm — named in an earlier version’s Limitations as deliberately untested — returned BODY-FITTED-BOUND-OPEN against it. Both rules, both thresholds and both verdict strings were committed to a public repository before the runs that read them. One implementation detail was not: the criterion that excludes the nodes with no wall-normal coordinate was derived from a failed round-trip gate, written before any body-fitted verdict number existed, and committed after a two-case smoke test while the 200-case run was executing, with the rule and thresholds unchanged. The manuscript states that provenance in those terms. We regard this history as the strongest evidence we can offer that the surviving claims were not selected after the fact, and we would rather the editor learn it from us than infer it.

Reproducibility. Every headline number maps to a committed script and result file through a per-claim reproduction map, with a manifest recording seeds, environment and SHA-256 hashes. The decisive comparison is gated rather than asserted. Before it could compute a body-fitted number, the implementation run in physical coordinates was required to reproduce the registered physical-coordinate bound, and did so at a worst relative difference of exactly $0.000e+00$. The two arms are then computed in one pass from one node array, so the coordinate map is the only difference between them; the frame has zero coordinate collisions among the scored nodes and a worst round-trip of $2.1e-13$ relative to each channel’s spread; and rescoreing the physical arm on the restricted node set moves it by 0.8%, so the restriction does not produce the result. The software is released as a CPU-first open-source package with a frozen I/O contract and an

AirfRANS loader; the snapshot matching this manuscript is release v1.1.0, permanently archived at Zenodo (DOI 10.5281/zenodo.21277928), which also serves as the deposited research data the journal’s data policy requires. Each decision rule in the paper — including the one that returned a negative verdict against our own preferred outcome — was registered in a script docstring and committed to a public repository before the run it governs, and those commits are identifiable in the history.

Preprint disclosure. An earlier preprint of this line of work is posted at arXiv:2607.10333 (currently v3, “NeuroForge: self-auditing neural CFD surrogates with calibrated physics-residual trust”). I flag the relationship explicitly because the title and framing differ enough that a search would otherwise raise a fair question. That version led with a calibrated trust layer built on a physics residual. The present manuscript is not a revision of it in the ordinary sense: the trust layer and the force-coefficient covariate analysis have been *removed*, the physics-residual audit is reduced to a single subsection, and the native-resolution and measure-dependence experiments described above are new and are now the paper’s spine. The removed material is intended for separate papers that will carry their own preprints. I will replace arXiv:2607.10333 with the present version so that the public record and this submission agree, and the separate material will be posted under its own identifiers rather than silently duplicated. Neither the present manuscript nor the separate material is under consideration at another journal.

Suggested reviewers are listed separately. We have deliberately not named anyone for the competence the paper most needs from outside machine learning — interpolation and remap error on unstructured meshes, and body-fitted coordinates — and would welcome the editor’s choice there. The manuscript is original, is not under consideration elsewhere, and both authors have approved this submission. We declare no competing interests and include the declaration of generative-AI use required by Elsevier policy, together with a CRediT contribution statement. We have selected the **subscription** publishing route.

Thank you for your consideration.

Sincerely,

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